

YouSpeed Technical Report: System Architecture, Jurisdiction Handling, Bundle Generation, and Lightweight Matching Benchmarks for Offline Smartphone Speed-Limit Assistance

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Abstract

This technical report documents the current YouSpeed system for offline smartphone speed-limit assistance. The system couples regional OpenStreetMap-derived bundles, on-device area-context lookup, bounded local map matching, and jurisdiction-aware warning logic across two mobile clients: the iPhone release-track application and the Android internal alpha. The report treats four connected questions. First, it documents the current local-first architecture, bundle contract, and mobile-client feature scope. Second, it characterizes regulation handling and maxspeed provenance, including a Germany-specific provenance decomposition and a cross-country explicit-tag analysis over 45 European extracts. Third, it describes the actual bundle-generation and iterative-update pipeline, including country selection, regional sharding, delta-pack publication, and the current v3 database schema. Fourth, it reports the benchmark evidence for four runtime data architectures, ten investigated matcher approaches on a retained hindsight corpus, and two deployable matcher profiles on closed-loop replay corpora. S3, a single-file spatially indexed database, remains the stable data-layer choice. In the offline matcher ladder, the three-way fixed-lag smoother attains the best accuracy and changed-example recall, while the current conservative matcher maximizes keep-current accuracy. In the deployable runtime comparison, the corridor-aware profile improves replay quality but increases bundle size, service-query latency, and oscillation counts, leaving the present size/latency-weighted product score slightly in favor of the smaller connected baseline. The current system is therefore asymmetric: the bundle and data architecture are stable, jurisdiction-aware warning logic is broader than jurisdiction-aware legal-default inference, and matcher selection remains an explicit trade-off between conservative stability and richer corridor state.

Keywords: intelligent speed assistance offline-first OpenStreetMap map matching mobile GIS smartphone systems

1 Introduction

Intelligent speed assistance (ISA) is now a regulatory requirement for new vehicle types in the European Union, yet the installed vehicle base remains far larger than annual new registrations [1; 2; 3; 4; 5]. This keeps the retrofit question open: practical speed-limit assistance still needs to run on devices that drivers already carry. Smartphone deployment is therefore relevant not as a theoretical alternative to integrated vehicle systems, but as a current path to broad availability.

YouSpeed addresses that path with a local runtime that keeps map data, area-context lookup, matching, and local correction handling on device. The current codebase spans a shared preprocessing and bundle-publication pipeline, an iPhone client on the launch path, and an Android internal alpha that follows the same bundle contract and runtime logic. The engineering question is no longer whether offline inference is possible in principle. The current question is which architecture and which lightweight matching approach best fit the present product and evidence state, while remaining honest about jurisdiction-specific regulation handling and update logistics.

This report reflects that current scope. It does not present a future architecture roadmap, and it does not treat unimplemented crowd synchronization as a central result. Instead, it documents the current system and compares the approaches that are active in the codebase today:

1. the implemented local-first system architecture and mobile-client feature scope,
2. the way the runtime currently handles jurisdiction identity, warning rules, and speed-limit provenance,
3. the bundle-generation and iterative-update machinery that turns country-scale OSM data into mobile artifacts,
4. four runtime data architectures (S1–S4) for local geospatial lookup,
5. a broader ladder of lightweight matching approaches on a retained hindsight benchmark,
6. two deployable matcher profiles on closed-loop replay corpora.

2 Related Work

Map matching has long been framed as a trade-off between geometric simplicity and sequence-aware robustness. The classic survey by Quddus et al. [6] organizes the field into geometric, topological, and probabilistic families. Sequence models then become the dominant answer to noisy and sparse trajectories: the HMM formulation of Newson and Krumm [7], ST-matching by Lou et al. [8], online HMM variants for streaming data [9; 10], low-latency path inference by Hunter et al. [11], and faster precomputed transition schemes such as FMM [12]. These methods demonstrate that later context is powerful, but they also assume a route-inference budget that is larger than the one targeted by the current YouSpeed runtime.

Smartphone-specific work confirms that this literature transfers to consumer sensing, but again usually in a richer probabilistic setting than the one used here. Bierlaire et al. [13] evaluate a probabilistic matcher directly on smartphone GPS traces, and production engines such as OSRM, Valhalla Meili, and Mapbox expose candidate radiuses, transition controls, and confidence-style outputs as first-class API concepts [14; 15; 16]. The present report borrows the scientific lesson that sequence context matters, while keeping the implementation target smaller and fully local.

Recent learning-based work pushes map matching further toward transfer learning, personalization, and adaptation under limited labels. Transformer transfer learning [17], deep inverse reinforcement learning with multi-intention modeling [18], preference-aware HMMs [19], and meta-learning approaches [20] all show that richer models can exploit driver- or context-specific structure. These lines are important context for YouSpeed, but they sit one layer above the current report. The present system still prioritizes bounded local state, auditable heuristics, and direct replay evaluation on device-shaped corpora.

The report also sits close to deployment-oriented packaging work rather than only to inference models. Single-archive spatial delivery formats such as MBTiles and PMTiles show that packaging choices are themselves a mature systems topic [21; 22]. That point matters because YouSpeed currently compares not only matching logic but also how map data is physically packaged, updated, and routed at country scale. The distinctive contribution of this report is therefore the joint treatment of lightweight matching, jurisdiction handling, mobile-client runtime scope, runtime data architecture, and bundle publication under present product constraints.

3 System Architecture

3.1 Current end-to-end architecture

The implemented YouSpeed path is local-first and offline-first. OSM snapshot ingestion and preprocessing run off device. The build pipeline emits regional v3 bundles that package road geometries, speed tags, area-context polygons, coverage metadata, and optional corridor tables in a single runtime artifact. Mobile clients load bundled target manifests, choose country or regional artifacts, verify checksums, and activate bundles atomically. During driving, each fix is routed to the best downloaded bundle through a coverage

bounding-box prefilter followed by coverage-polygon containment. The selected bundle then executes local lookup, speed derivation, and way selection without a route-level cloud dependency.

Figure 1 summarizes the current paper-scope architecture. The broader cloud corroboration stack that appears in internal architecture documents remains a north-star extension, not the present critical path. The evaluated critical path is snapshot-to-bundle publication, local bundle activation, on-device inference, and editor-mediated export of reviewed local observations.

The benchmark harness sits beside this runtime rather than replacing it. The S1–S4 latency benchmark reuses the same Germany-derived artifacts and query primitives but isolates spatial lookup and area-context retrieval at one fixed probe point. The replay benchmarks reuse the same runtime matcher implementations and logs, but they remove the user-interface lifecycle so that candidate arbitration, tunnel handling, and oscillation behavior can be measured directly.

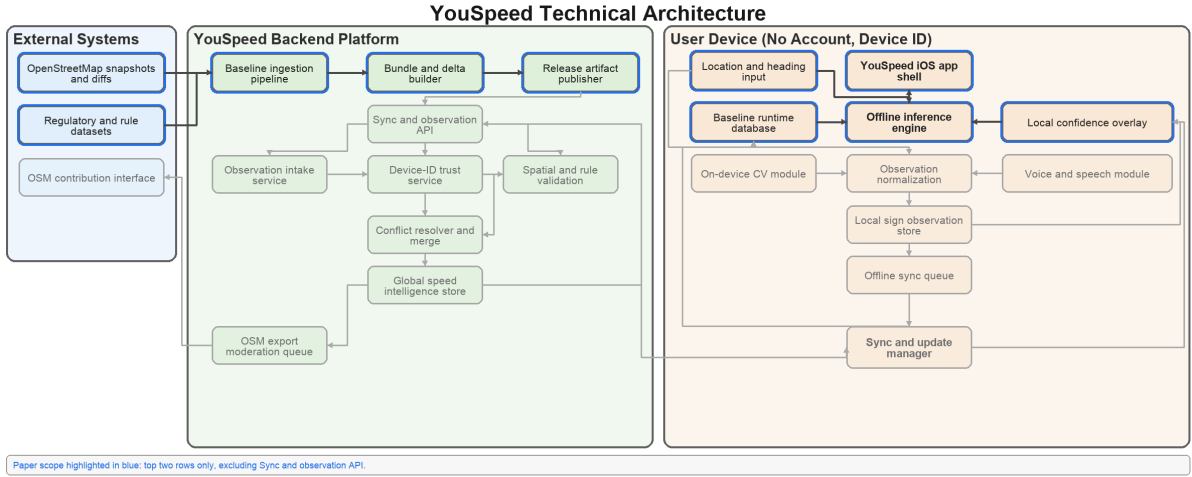


Figure 1: Current paper-scope system architecture. The implemented critical path is local-first: regional bundle publication, on-device inference, and editor-mediated observation export.

3.2 iPhone application

The iPhone application is the current release-track client. It bootstraps from a bundled Karlsruhe seed, discovers Germany shard manifests from the bundled `BundleTargets.top10.json` contract, downloads full bundles or verified SQL delta chains when available, assembles multipart database payloads when needed, and activates bundles atomically. The runtime uses plain `SQLite3` rather than runtime `mod_spatialite` loading, with an RTree bounding-box prefilter, polyline distance refinement, optional heading-aware scoring, deterministic tunnel handling, and bundle-coverage routing across downloaded regions.

The same client also carries the current driver-facing feature surface. It presents current speed, inferred limit, street and city context, tunnel state, and penalty-aware overspeed warnings. It captures local speed-limit observations through an on-device speech-recognition workflow, stores them in a review queue, and exports approved proposals as editor-oriented `.osc` packages rather than publishing directly from the app. The iPhone codebase is also the main source of inspector logs and replay corpora used in the matching benchmarks reported later in this report.

3.3 Android application

The Android application is an internal alpha that follows the same bundle and manifest contract as the iPhone client, but it is not the current public release gate. It bootstraps from a bundled seed, derives the same Germany shard endpoints, verifies and activates full bundles or multipart bundle payloads, and mirrors the same local lookup contract. The Android matcher adopts the same bounded-history logic family as the iPhone runtime: same-reference promotion, linked-way continuity, same-reference bounce

suppression, tunnel-portal continuity, and corridor-state carryover. It also includes a fallback path for Android builds that expose stored RTree tables without exposing the SQLite RTree module at runtime.

The Android alpha emphasizes parity experiments that are useful for architecture validation. It adds spoken overspeed and driving-ban warnings, a bundled German offline Vosk recognizer for local speed capture, SQLite-backed review and export of local observations, debug log export, OSM browser handoff, and emulator or device instrumentation for replay and live bundle bootstrap. This makes Android scientifically relevant even before public launch because it exercises the shared bundle contract and runtime logic under a second mobile platform.

Table 1: Current mobile-client feature scope.

Capability	iPhone release-track app	Android internal alpha
Bundle lifecycle	Bundled seed, Germany shard discovery, full-bundle or SQL-delta updates, multipart assembly, atomic activation.	Bundled seed, shared shard discovery, full-bundle bootstrap, multipart assembly, atomic activation.
Lookup and matching	S3 SQLite3 lookup, heading-aware ranking, coverage-based regional routing, deterministic tunnel handling.	Shared v3 lookup contract, iPhone-shaped continuity logic, corridor-state carryover, bbox fallback when Android lacks live RTree support.
Warning surface	Driver-facing speed or limit UI, tunnel state, penalty-aware overspeed presentation, haptic and speech driving-ban alerts.	Compose parity UI, spoken overspeed and driving-ban alerts, tunnel and matcher debug state.
Local observations	On-device speech capture, review queue, single or bulk .osc editor export.	Bundled Vosk offline speech capture, SQLite review queue, single or bulk .osc export, OSM browser handoff.
Verification	Unit tests, route replay, tunnel regressions, geometry-stress analysis on iPhone logs.	Emulator or device instrumentation, live Germany-shard bootstrap test, replay regressions, runtime fallback checks.

4 Jurisdiction Handling and Maxspeed Provenance

4.1 Separation of bundle identity, speed inference, and warning semantics

The current system handles jurisdiction-specific behavior in layers rather than through one monolithic legal model. At bundle level, the manifest can carry a three-letter `country_code`, optional coverage metadata, and an optional country-specific penalty-rule artifact. At inference level, the way table stores explicit `maxspeed`, inherited `maxspeed:type`, and inherited `source:maxspeed` strings alongside road class and area-context inputs. At consumer-warning level, the app loads country-specific penalty-rule JSON that can encode different currencies, languages, locality variants, penalty points, and conditional driving-ban logic. This separation makes the runtime robust to incomplete map tagging, but it also means that jurisdiction handling is broader in the warning layer than in the unsigned-road legal-default layer.

Table 2: Current jurisdiction-handling layers in the runtime.

Layer	Mechanism	Current scope and limitation
Bundle identity	Manifest <code>country_code</code> , coverage bbox or polygon, country bundle catalog.	Multi-country and multi-region routing is supported in the artifact contract.
Speed-tag ingestion	<code>maxspeed</code> , <code>maxspeed:type</code> , <code>source:maxspeed</code> , road class, tunnel tag, service tag.	Works wherever explicit or inherited tags exist.
Locality split	<code>insideCity</code> derived from way-class precedence first, then residential polygons, then city-context fallback.	Supports inner/outer locality warnings, but the legal-default use is strongest for Germany.
Warning semantics	Country-specific penalty JSON with currency, language, bands, locality variants, and conditional driving-ban metadata.	Present in both mobile apps and broader than the current legal-default inference layer.
Unsigned-road defaults	Inherited <code>urban/rural/motorway</code> parsing, then road-class heuristics with residential override.	Implemented locally, but current semantics are still validated mainly for Germany.

Two current examples show how this split works in practice. The German ruleset encodes separate inner-city and outer-city penalty variants and conditional driving bans for repeat offenses. The Netherlands ruleset uses simpler national bands without a locality split, but it still carries jurisdiction-specific currency,

source metadata, and high-overspeed conditions. The runtime resolves these rule differences after overspeed estimation through `country_code` and `insideCity`, not by changing the map-matching core itself.

The currently packaged rule portfolio spans ten jurisdictions: Belgium, Germany, Great Britain, Iceland, Liechtenstein, Luxembourg, Monaco, the Netherlands, Romania, and Sweden. This means that jurisdiction-specific warning semantics already cover a broader set of countries than the present country-specific legal-default inference layer.

The current speed-derivation logic is more conservative. Explicit numeric `maxspeed=*` tags take precedence. If they are absent, the runtime checks inherited tags for strings containing `urban`, `rural`, or `motorway`. If neither explicit nor inherited tags are decisive, it falls back to road-class heuristics and then optionally overrides those heuristics with the current `insideCity` decision. This logic keeps the code auditable and fast, but it is not yet a full country-by-country legal-default engine.

4.2 Germany provenance decomposition

All evaluated artifacts derive from the same Germany OSM snapshot (`germany-latest.osm.pbf`, 4.70 GB) [23]. Table 3 summarizes the current Germany dataset profile and the provenance decomposition that the runtime sees after rule-aware fusion.

Table 3: Germany dataset profile and speed-limit provenance used by the current runtime.

Metric	Value
Input file size	4.70 GB
Car-drivable ways	7,963,481
Ways with <code>maxspeed=*</code>	2,637,020
Ways with <code>zone:maxspeed=*</code>	172,116
Ways with explicit speed-limit tag	2,639,693
Area-context objects	129,459
Explicit tagged limit share	33.15%
Implicit regulatory tag share	0.98%
Context-based legal fallback share	65.87%

The Germany profile is the central reason the report keeps provenance explicit. Roughly one third of car-drivable ways already carry explicit speed limits in OSM. Another 0.98% contribute inherited regulatory context through rule-bearing tags. The remaining 65.87% depend on default interpretation or area context. That asymmetry is exactly why the product has to couple map matching with area-context lookup and why pure explicit-tag retrieval would be insufficient even before matching errors are considered.

The current area-context logic is also deliberately narrower than administrative-jurisdiction modeling. For limit inference, the runtime first uses way-class precedence (`residential`, `service`, `crossing`, `living_street`) to assert built-up context where appropriate. If that does not decide the case, it resolves residential polygons. Only then does it use broader city context as a fallback signal. This means that `insideCity` in the current runtime is a built-up-driving proxy, not a general administrative-boundary classifier.

4.3 Cross-country explicit-tag statistics

The repository also contains a Europe-wide comparative statistic over 45 country extracts. This analysis counts car-drivable ways and explicit `maxspeed=*` coverage per country, using the same drivable-highway definition as the bundle generator. It is therefore a cross-country provenance statistic for explicit-tag availability, not a full country-by-country decomposition into explicit, inherited, and fallback classes.

The heterogeneity is substantial. The explicit-share range spans from 3.10% (Turkey) to 67.75% (Netherlands), with a median of only 16.15%. Germany sits above both the European median and the European weighted mean, but it is still far from the top of the distribution. This matters for system design: a country portfolio chosen purely by map size would not necessarily align with a country portfolio chosen by explicit-tag completeness.

Table 4: Descriptive statistics for explicit `maxspeed=*` share across 45 European extracts.

Statistic	Value
Countries analyzed	45
Minimum country share	3.10%
First quartile	13.11%
Median	16.15%
Arithmetic mean	20.83%
Way-weighted mean	22.61%
Third quartile	25.35%
Maximum country share	67.75%
Germany rank in Europe-wide ordering	8
Germany explicit-tag share	33.12%

Table 5: Current bundle-target countries and their explicit `maxspeed=*` coverage.

Rank	Country	ISO2	Share	Bundle mode
1	Netherlands	NL	67.75%	single country
2	Romania	RO	62.55%	single country
3	Luxembourg	LU	53.60%	single country
4	Belgium	BE	37.06%	single country
5	Sweden	SE	35.89%	single country
6	Liechtenstein	LI	33.96%	single country
7	Iceland	IS	33.78%	single country
8	Germany	DE	33.12%	regional shards
9	Monaco	MC	29.00%	single country

The current bundled target set is therefore not arbitrary. Its mean explicit-tag share is 42.97%, compared with 20.83% across all 45 European extracts, and its way-weighted mean is 39.25%. In other words, the current target set clusters well above the continental baseline for explicit-tag availability. This does not remove the need for fallback logic, but it does reduce the probability that the initial rollout set is dominated by very low-explicitness jurisdictions.

At the same time, the report should be clear about what this comparative statistic does not yet provide. The Europe-wide analysis currently measures explicit-tag prevalence, and the Germany snapshot already carries a fuller provenance split because its fallback semantics are the best validated in the current codebase. Extending the same explicit or inherited or fallback decomposition to every target country is possible, but it is not yet the standard comparative report generated by the current toolchain.

5 Bundle Generation and Iterative Updates

5.1 Country selection and regional sharding

Bundle generation starts from comparative data availability, not from manual country selection alone. The script `analyze_europe_maxspeed_share.py` scans European Geofabrik extracts, counts car-drivable ways, and produces the cross-country explicit-tag ranking discussed above. The top-ranked countries feed the bundle-target configuration used by the apps. The generator then decides whether a target stays a single-country bundle or is split into child regions.

That split is governed by a simple operational rule: if a country PBF stays at or below a 1 GB threshold, it remains a single bundle; if it exceeds the threshold, the planner switches to Geofabrik child regions. Germany therefore becomes a 16-region shard set in the current target file, while smaller countries such as the Netherlands or Luxembourg remain single-country bundles. This rule does not optimize every possible deployment objective, but it is transparent, reproducible, and aligned with mobile download constraints.

5.2 SQLite bundle construction

After target selection, `build_spatialite_v3.py` constructs the current S3 runtime artifact. The build starts from the v1 dist representation, creates the v3 SQLite database, loads car-drivable ways with their speed-relevant attributes, writes RTree bounding boxes, stores polyline geometries, and copies polygonal context needed for residential inference. Optional topology and corridor stages then add a detailed shared-endpoint graph and paired corridor tables for tunnel and motorway continuity.

The current corridor-aware Karlsruhe bundle illustrates the resulting artifact scale. Its latest research bundle contains 242,359 way rows, 32,872 area rows, 435,372 detailed way-link rows, 11,961 corridor-progress rows, and 1,656 corridor-pair rows. Metadata in the same bundle records the spatial back-end (`sqlite_rtree`), the detailed link mode (`shared_endpoint_detailed`), and the corridor mode (`paired_portal_chain_v1`). This is the bundle family that underlies the corridor-aware replay numbers reported later.

5.3 Publication artifacts and delivery contract

The builder and publisher then transform the SQLite runtime database into app-consumable release artifacts. Table 6 summarizes the current stages.

Table 6: Current bundle-generation and publication stages.

Stage	Main script	Primary output	Role
Comparative scan	<code>analyze_europe_maxspeed_share.py</code>	Europe ranking CSV and markdown	Country prioritization by explicit-tag coverage.
Sharding plan	<code>plan_country_region_bundles.py</code>	Country bundle-plan JSON	Decide single-country vs regional shards.
Bundle build	<code>build_spatialite_v3.py</code>	<code>speeds_v3.sqlite</code>	Create current S3 runtime database.
Manifest publish	<code>publish_v3_bundle.py</code>	Bundle directory plus manifest	Attach db artifact, optional split parts, coverage, penalty rules, and delta index.
Country catalog	<code>build_v3_country_bundle_catalog.py</code>	Country catalog JSON	Aggregate regional manifests for country-level discovery.
Target config	<code>generate_v3_country_bundles.py</code>	<code>BundleTargets.top10.json</code>	Ship the current mobile download target list.

The bundle manifest is deliberately richer than the minimal local file copy. It can reference one full database, an optional list of split `db_parts`, an optional delta index, an optional penalty-rule JSON, and optional coverage metadata. If a bundle exceeds the release-asset threshold, the publisher splits the database into numbered parts while keeping one logical `db` payload in the manifest for identity and checksum semantics. Country catalogs then aggregate multiple regional manifests and their coverage metadata for bundle routing on device.

5.4 Iterative updates

The iterative-update path is already implemented, even though the current release-track behavior remains conservative. The update pipeline starts from daily Geofabrik replication diffs. `build_v3_delta_pack.py` can either infer a patch directly from an `.osc(.gz)` diff plus the base database or build an exact SQL patch by diffing a rebuilt target database against the base database. The generated delta manifest records the from-version, to-version, patch artifact, and patch statistics. `roll_v3_delta_index.py` then merges new delta manifests into a rolling index with the current retention window of 30 entries.

This update path is not only specified; it is also measured. The current daily-diff analysis covers a 30-day Germany window from 2026-01-22 to 2026-02-20. Table 7 summarizes the workload seen by the update machinery.

Table 7: Germany daily-diff update statistics over the current 30-day analysis window.

Metric	Mean	Median	Max
Changed ways per day	60,972	61,076	69,793
Maxspeed-tag changes per day	4,667	4,643	5,668
Invalidated S1 tiles	6,712	6,728	7,756
Invalidated S2 tiles	2,685	2,684	2,907
V3 SQL patch rows	134,965	131,253	158,028
V3 patch apply time (ms)	1,024	1,026	1,180
V4 SQL patch rows	182,929	177,938	213,766
V4 patch apply time (ms)	1,357	1,359	1,581

Three observations follow from these data. First, day-to-day map maintenance is not sparse at country scale: the median day still moves more than 61,000 ways and more than 4,600 speed-related tags. Second, the update burden is architecture-dependent. The same diff window invalidates roughly 2.5 times as many S1 tiles as S2 tiles, which is one reason why coarse global indexing remains unattractive for mobile delivery. Third, exact SQL patch application is already operationally plausible for S3-sized artifacts in the current analysis, with a median simulated apply time of about 1.03 s. This does not yet prove that every mobile update path is production-ready, but it does move iterative updates from abstract future work into an implemented and measured subsystem.

6 Data and Methods

6.1 Runtime architecture scenarios

The four runtime architectures are summarized in Table 8. The benchmark keeps source data and scoring fixed so that architectural differences arise from packaging and spatial access, not from different road subsets or different ranking rules.

Table 8: Current runtime architecture scenarios evaluated in this report.

Scenario	Runtime artifact	Query characteristics
S1	Global country-scale index files	Broad candidate scans with high IO and weak locality.
S2	Content-addressed spatial tile packs	Localized reads and bounded regional fan-out.
S3	Single-file spatially indexed embedded database	Direct spatial-window retrieval with low deployment overhead.
S4	Embedded database with tile-window pre-filter	Two-stage pruning for dense candidate regions.

6.2 Investigated matching approaches

All investigated matchers operate on the same local candidate sets. For each fix t , candidate ways $i \in C_t$ receive a geometry score

$$s_{\text{geom}}(t, i) = \begin{cases} d_{t,i} + \lambda_h \cdot \Delta\theta_{t,i}, & \text{if course is reliable} \\ d_{t,i}, & \text{otherwise} \end{cases}$$

where $d_{t,i}$ is point-to-way distance and $\Delta\theta_{t,i}$ is bidirectional heading mismatch. Higher-level approaches then add bounded continuity state, short-history sequence logic, or lightweight learned ranking on top of this shared local candidate generation.

The matching study uses two complementary experimental settings. The retained offline benchmark fixes candidate windows from non-duplicate inspector logs and asks whether a model recovers hindsight labels. This benchmark isolates arbitration quality and makes it possible to compare multiple approaches that are not all deployed end to end. The current ladder spans four families:

- **Purely local heuristics:** the lowest-distance baseline and the current conservative final matcher.
- **Short-history sequence heuristics:** the mini-HMM else heuristic.
- **Bounded-history decision and smoothing models:** two-way tree gates, three-way tree gates, and a three-way fixed-lag smoother.
- **Lightweight learned rankers:** tree, logistic, tiny-MLP, and online-linear candidate rankers.

The closed-loop replay benchmark is narrower and more deployment-oriented. It keeps bundle queries, session state, tunnel logic, and runtime size or latency terms, and therefore compares only two deployable S3 profiles on the same data layer:

- **Connected baseline.** This profile uses exact geometry distance, heading-aware scoring, connected-way continuity, and tunnel or motorway entry constraints derived from a minimal shared-endpoint graph. It does not materialize explicit corridor tables.
- **Corridor-aware profile.** This profile uses the same geometric core but adds corridor identity, corridor phase, paired entry and exit chains, and more detailed shared-node topology. It activates the runtime three-way gate whenever corridor-aware arbitration is useful.

6.3 Query modes and benchmark harness

All architecture scenarios use the same query modes:

- **bbox:** distance to the clamped point on a candidate bounding box,
- **polyline:** minimum point-to-segment distance over full geometry,
- **hybrid:** bbox pre-ranking followed by polyline refinement of a small top- N set,
- **polycontainment:** residential-polygon area-context lookup for built-up classification.

The architecture benchmark uses a fixed Berlin probe point (52.5200, 13.4050), heading 90°, and $k = 5$ candidates. It runs in two contexts: host-side execution on a MacBook Pro (Apple M4 Max, 48 GB RAM, macOS 15.5) and on-device execution on an iPhone 14 Pro (iOS 26.2.1). This harness isolates spatial lookup and area-context retrieval. It does not reproduce the full recurrent app runtime and it does not currently benchmark Android device latency.

6.4 Replay corpora and metrics

The current matcher comparison uses three corpora with different purposes:

Table 9: Current evaluation corpora.

Corpus		Purpose	Logs	Fixes	Hindsight labels
Retained	offline	Candidate-level hindsight benchmark	7	9,636	5,513
benchmark		from non-duplicate inspector logs			
Inspector replay	corpus	Closed-loop replay of current runtime behavior on field logs	10	11,351	6,529
Geometry-stress	corpus	Dedicated tunnel and same-ref stress replay	2	2,598	1,410

The retained offline benchmark contains 5,513 hindsight-labelled examples, including 381 switch-labelled cases, and it uses 3,307/1,103/1,103 train/validation/test examples where supervised models require a learned split. For profile comparison, the primary metrics are replay accuracy a , changed-example recall r , and unchanged-example accuracy u . To summarize replay quality without collapsing everything into one number too early, we first use

$$A = 0.5a + 0.25r + 0.25u.$$

Because the current product still cares about bundle size and service latency, we also report the present internal score

$$S = 100 \left(0.7A + 0.2 \frac{t_{\min}}{t} + 0.1 \frac{b_{\min}}{b} \right),$$

where t is service-query $p95$ latency, b is bundle size, and $t_{\min}$, $b_{\min}$ are the best values among compared profiles. We treat S as a product-weighted summary, not as a universal scientific metric.

7 Results

7.1 Runtime architecture comparison

Table 10 reports the joint latency matrix, and Figure 2 shows the same values grouped by scenario.

Table 10: Joint host/device latency matrix at the benchmark coordinate (ms).

Execution	Mode	S1	S2	S3	S4
host	bbox	5451.24	172.21	64.74	166.26
host	hybrid	10152.12	169.25	29.56	64.00
host	polyline	10300.65	170.65	38.88	72.03
host	polycontainment	1.51	1.07	0.94	0.44
device	bbox	4526.75	232.58	43.77	792.13
device	hybrid	2935.52	61.03	24.10	99.57
device	polyline	3122.36	76.20	39.16	78.54
device	polycontainment	3.14	0.81	0.74	2.79

The architecture result is clear. S1 is not competitive for country-scale mobile inference. S2 is a substantial improvement and keeps the locality benefits of explicit tiling, but its high file count remains operationally heavier than the single-file variants. S4 does not justify its extra prefilter stage under the current workload; it loses to S3 on device in every decision-relevant query mode. S3 therefore remains the stable data-layer choice in the current project state.

The area-context result is also narrow and current-scope only. The measured `polycontainment` step is a residential-polygon lookup used by the benchmark harness for built-up classification; it is not an administrative-boundary benchmark. Under that definition, S3 remains fastest on device at 0.74 ms.

7.2 Offline matcher benchmark across investigated approaches

Table 11 reports the retained offline benchmark across the ten investigated approaches. This benchmark isolates candidate arbitration rather than end-to-end runtime behavior, so its main value lies in comparing approach families under a fixed candidate corpus.

Three patterns emerge from this ladder. First, bounded future or sequence context matters. The three-way fixed-lag smoother reaches the strongest overall operating point at 94.20% accuracy and 55.34% changed-example recall, while the mini-HMM and tree-gated approaches also improve change recovery relative to the current conservative matcher. Second, the current conservative matcher should be interpreted as a stability reference rather than as a general optimum: it maximizes keep-current accuracy at 100.00%, but it does so by suppressing all hindsight-labelled switches in this benchmark. Third, the current label volume does not yet make the small learned rankers dominant. The tree candidate ranker is competitive,

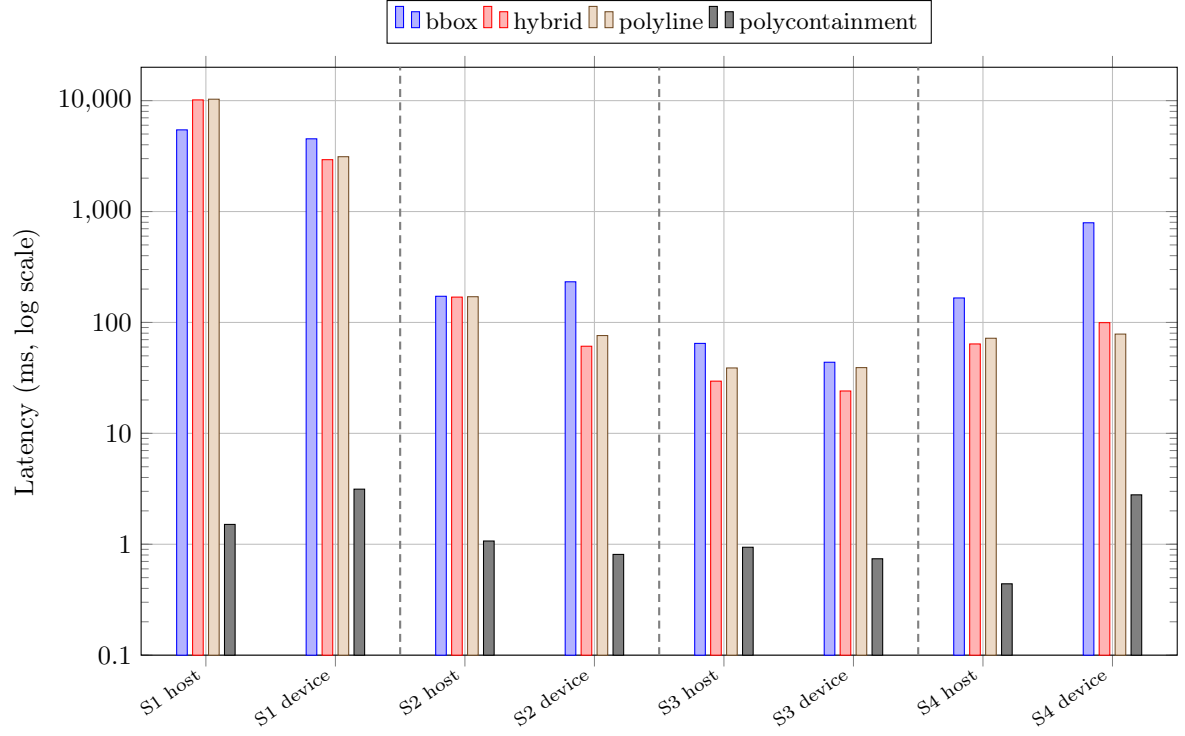


Figure 2: Scenario-first latency view for the current S1–S4 comparison.

Table 11: Retained offline benchmark across investigated matcher approaches (9,636fixes, 5,513hindsight labels).

Approach	Accuracy	Changed rec.	Keep-current acc.
Current final matcher	90.66%	0.00%	100.00%
Mini-HMM else heuristic	92.57%	34.95%	98.50%
Lowest-distance baseline	90.75%	54.37%	94.50%
Two-way tree gate	93.56%	46.60%	98.40%
Three-way tree gate	93.11%	42.72%	98.30%
Three-way fixed-lag smoother	94.20%	55.34%	98.20%
Tree candidate ranker	93.11%	50.49%	97.50%
Logistic ranker	91.02%	41.75%	96.10%
Tiny MLP ranker	87.04%	20.39%	93.90%
Online linear ranker	90.75%	48.54%	95.10%

but logistic, online-linear, and especially tiny-MLP variants remain below the stronger bounded-history heuristic and smoother baselines.

The ladder is also scientifically useful because it separates different notions of improvement. The lowest-distance baseline already reaches 54.37% changed-example recall, which means that many switch-labelled cases are geometrically recoverable without rich temporal state. The gain from more structured models therefore lies less in discovering switches at all than in balancing switch recovery against keep-current stability. On the current retained corpus, the fixed-lag smoother is the best compromise inside that balance.

7.3 Current runtime matcher profile comparison

The closed-loop runtime comparison narrows the field to two deployable S3 profiles and adds runtime costs that the offline ladder does not measure: bundle size, service latency, gate activity, and geometry-log oscillation behavior. Table 12 compares these two profiles on replay and geometry-stress corpora.

Table 12: Current matcher profile comparison on replay and geometry-stress corpora.

Profile	Bundle (MiB)	Service p95	Replay acc.	Changed rec.	Unchanged acc.	Geom agree.	Way ABA	Score S
Connected base-line	108.5	0.132	93.41%	49.23%	96.71%	80.45%	4	88.23
Corridor-aware	128.2	0.166	95.99%	72.19%	97.76%	83.56%	89	87.73

The corridor-aware profile is not a trivial strict improvement. It is better on replay quality: accuracy rises by 2.58 percentage points, changed-example recall rises by 22.96 percentage points, unchanged-example accuracy also improves, and geometry-log agreement is higher. The same profile is also materially more active, invoking the current three-way gate 3,118 times across the replay corpus.

At the same time, the connected baseline keeps two practical advantages. It is smaller by 19.7 MiB and faster at service-query $p95$, and it shows much less ABA behavior in the geometry-stress logs. Under the present product-weighted score S , these latency and size terms are enough to move the aggregate result slightly back toward the baseline (88.23 vs. 87.73), even though the replay-accuracy composite A favors the corridor-aware profile (0.8319 vs. 0.9048). The current scientific conclusion is therefore a trade-off statement, not a winner-takes-all statement.

This trade-off is visible in the corpora details. The replay corpus contains 934 motorway fixes, 92 motorway-link fixes, and 187 selected tunnel fixes. The geometry-stress corpus contains 82 replay tunnel fixes and one portal-eligible tunnel case. These are exactly the route types for which extra corridor state is expected to matter. The present data confirm that expectation on replay accuracy, but they also show that richer corridor state can reintroduce oscillation pressure in dedicated stress logs.

8 Discussion

The current report separates four questions that were previously mixed together.

First, the architecture question is now relatively stable. S3 is the strongest present data-layer choice because it combines the lowest device-side latency with single-file deployment and no extra prefilter stage. The iPhone and Android clients both validate this architectural center of gravity through the same regional bundle contract, even though the iPhone client is the release-track product surface and Android remains an internal alpha.

Second, the jurisdiction question is layered rather than binary. The bundle and warning layers already carry explicit country identity and country-specific penalty semantics. The map-inference layer already handles explicit and inherited speed tags in a country-agnostic way where the tags themselves are informative. But unsigned-road defaults still rely on a more limited heuristic stack whose best-validated semantics remain German. That asymmetry is not a flaw in the description; it is the actual current system state.

Third, the update question is no longer hypothetical. Comparative target-country ranking, sharding, exact delta-pack generation, rolling delta indexes, and manifest publication are all implemented, and the

Germany 30-day update study shows that these subsystems operate on realistic country-scale workloads. What remains open is production confidence across every target and every mobile download path, not whether an iterative-update mechanism exists at all.

Fourth, the matching question is best understood on two levels rather than one. In the retained offline benchmark, added bounded future context pays off: the fixed-lag smoother and tree-gated approaches outperform purely local conservative arbitration on changed-example recovery, and the best lightweight learned models still trail the strongest bounded-history non-neural baselines. In the closed-loop runtime benchmark, the comparison becomes narrower because deployable profiles must also pay for bundle size, service latency, and route-stability side effects.

9 Future Work

Several next steps follow directly from the current evidence.

The first concerns jurisdiction-aware defaults. The present bundle contract and warning layer already scale across jurisdictions, but the legal-default inference layer still deserves a fuller country-by-country model. Extending the explicit or inherited or fallback provenance decomposition beyond Germany is therefore not just a reporting improvement; it is also the measurement foundation for more principled country-specific default handling.

The second concerns iterative updates. The exact delta-pack machinery is implemented and measured, but the rollout path remains intentionally conservative. A natural next step is to move from the Karlsruhe seed-first validation pattern to a broader country and shard matrix, with stronger end-to-end evidence for multipart downloads, stale-version recovery, and repeated patch application on real devices rather than only host-side simulation and narrow mobile-path verification.

The third concerns matcher convergence. The current results show a productive tension between conservative baseline behavior and corridor-aware replay gains. Future work should therefore evaluate richer profiles not only on replay accuracy but also jointly with bundle build cost, delta size, update friction, and geometry-stress stability, so that matcher adoption reflects the full operational envelope of the product.

10 Conclusion

The current YouSpeed system is best described as a four-layer result. At the architecture layer, a local-first bundle pipeline and S3 runtime data layer define the stable implementation center. At the client layer, the iPhone app provides the current launch-facing runtime while the Android alpha validates the same contract and matcher family on a second mobile platform. At the regulation layer, country identity and penalty semantics are already decoupled from the matcher, while unsigned-road legal defaults remain strongest for Germany. At the matcher layer, the project has multiple defensible operating points: the fixed-lag smoother is strongest in the retained offline arbitration benchmark, while the deployable runtime comparison remains a trade-off between a smaller connected baseline and a richer corridor-aware profile with better replay quality. The scientific value of the current status lies in keeping these comparisons explicit instead of forcing them into a single winner narrative.

Reproducibility

All preprocessing, scenario construction, benchmark steps, bundle-generation stages, and update scripts are parameterized in the accompanying codebase at <https://github.com/volzinnovation/youspeed.de>. The current report ties its measurements to the Germany snapshot dated 2026-02-23, the Europe-wide explicit-tag analysis artifacts stored under `mapdata/reports/`, the 30-day Germany daily-diff analysis window from 2026-01-22 to 2026-02-20, the fixed Berlin probe benchmark, and the replay corpora summarized in Table 9.

A Bundle Artifact Contracts

The runtime artifact family uses three JSON contracts: the bundle manifest, the delta manifest, and the country catalog. Table 13 summarizes the bundle manifest fields. Table 14 summarizes the delta manifest fields. Table 15 summarizes the country catalog fields.

Table 13: Current v3 bundle-manifest fields.

Field	Required	Meaning
<code>format</code>	yes	Constant identifier <code>youspeed.v3.bundle.manifest</code> .
<code>schema_version</code>	yes	Manifest schema version.
<code>variant</code>	yes	Runtime family identifier, currently v3.
<code>region</code>	yes	Logical bundle region id.
<code>country_code</code>	no	ISO alpha-3 jurisdiction code used for bundle identity and rule selection.
<code>bundle_version</code>	yes	Logical bundle version string.
<code>created_at_utc</code>	yes	Manifest creation timestamp.
<code>min_app_version</code>	yes	Minimum compatible app version.
<code>db</code>	yes	Logical database artifact with file name, bytes, checksum, and optional URL.
<code>db_parts</code>	no	Optional list of split database parts for release-asset size limits.
<code>delta_index</code>	no	Optional rolling index of available incremental updates.
<code>penalty_rules</code>	no	Optional country-specific penalty-rule artifact.
<code>coverage</code>	no	Optional bbox plus optional polygon artifact for on-device bundle routing.
<code>self</code>	no	Manifest file identity and optional published URL.

Table 14: Current v3 delta-manifest fields.

Field	Required	Meaning
<code>format</code>	yes	Constant identifier <code>youspeed.v3.delta.manifest</code> .
<code>schema_version</code>	yes	Delta schema version.
<code>region</code>	yes	Logical region id covered by the patch.
<code>from_bundle_version</code>	yes	Source bundle version for the patch.
<code>to_bundle_version</code>	yes	Target bundle version for the patch.
<code>created_at_utc</code>	yes	Delta-manifest creation timestamp.
<code>source_diff_file</code>	no	Input replication diff used to generate the patch.
<code>patch</code>	yes	SQL patch artifact with file, bytes, checksum, URL, and compression metadata.
<code>stats</code>	yes	Patch statistics including changed ways, speed-tag changes, and SQL row counts.

Table 15: Current v3 country-catalog fields.

Field	Required	Meaning
<code>format</code>	yes	Constant identifier <code>youspeed.v3.bundle.catalog</code> .
<code>schema_version</code>	yes	Catalog schema version.
<code>variant</code>	yes	Runtime family identifier, currently v3.
<code>country</code>	yes	Country identifier for the catalog owner.
<code>bundle_version</code>	yes	Logical catalog version.
<code>created_at_utc</code>	yes	Catalog creation timestamp.
<code>max_country_pbf_bytes</code>	yes	Planning threshold that explains why sharding is or is not used.
<code>regions</code>	yes	List of region entries with manifest artifact and optional coverage metadata.

B Current v3 Bundle Database Schema

Table 16 documents the schema of the current corridor-aware Karlsruhe v3 bundle (`bundle_version 2026-03-11-shared-node-corridor-hmm`). Internal RTree shadow tables are omitted for readability. The current builder can also materialize additional intermediate tables such as `way_endpoints` during construction, but the deployed bundle documented here retains the tables that the runtime actually consumes.

Table 16: Current deployed v3 bundle schema and row counts (Karlsruhe latest corridor-aware bundle).

Table	Columns	Purpose	Rows
metadata	key, value	Build metadata, feature flags, source lineage.	8
ways	way_id, class, name or ref, explicit or inherited speed fields, heading, service, tunnel, bbox	Per-way speed, class, heading, and bounding-box attributes.	242,359
ways_rtree	way_id plus bbox coordinates	Spatial pruning for candidate way retrieval.	242,359
way_geom	way_id, points_json	Polyline geometry for point-to-segment refinement.	242,359
areas	row_id, area id, geometry or place fields, residential flag, geometry JSON, bbox	Residential polygons plus place or boundary context.	32,872
areas_rtree	row_id plus bbox coordinates	Spatial pruning for polygon and area-context lookup.	32,872
way_links	way_id, linked way id, shared ref, shared node key	Detailed shared-node continuity graph.	435,372
corridor_progress	Corridor kind or id, side node key, way id, depth and span measures	Stateful tunnel and motorway corridor arbitration.	11,961
corridor_pairs	Corridor id plus paired corridor identity fields	Pair entry and exit corridor components.	1,656

C Europe-wide Explicit Maxspeed Ranking

Table 17 reproduces the full current Europe-wide ranking used by the comparative provenance scan. The statistic measures explicit `maxspeed=*` coverage on car-drivable ways and is included here in full because it directly informs target-country prioritization and cross-country provenance interpretation.

Table 17: Europe-wide ranking by explicit `maxspeed=*` share on car-drivable ways.

Rank	Country	ISO2	Drivable ways	Tagged ways	Share (%)
1	Netherlands	NL	1,360,101	921,426	67.75
2	Romania	RO	671,248	419,884	62.55
3	Luxembourg	LU	57,401	30,766	53.60
4	Belgium	BE	816,110	302,452	37.06
5	Sweden	SE	1,235,544	443,443	35.89
6	Liechtenstein	LI	5,407	1,836	33.96
7	Iceland	IS	61,433	20,751	33.78
8	Germany	DE	7,964,480	2,637,436	33.12
9	Monaco	MC	1,224	355	29.00
10	United Kingdom	GB	4,871,056	1,275,216	26.18
11	Andorra	AD	3,714	955	25.71
12	Switzerland	CH	862,842	218,757	25.35
13	Hungary	HU	514,402	126,714	24.63
14	Austria	AT	1,257,719	287,600	22.87
15	Czech Republic	CZ	860,607	187,689	21.81
16	France	FR	7,111,322	1,490,044	20.95
17	Serbia	RS	381,641	77,588	20.33
18	Slovakia	SK	367,681	74,103	20.15
19	Malta	MT	26,092	4,658	17.85
20	Ireland and Northern Ireland	IE	1,096,604	195,628	17.84
21	Denmark	DK	886,408	157,710	17.79
22	Finland	FI	996,944	167,987	16.85
23	Spain	ES	2,720,041	439,358	16.15
24	Belarus	BY	534,645	85,966	16.08
25	Croatia	HR	298,960	47,342	15.84
26	Norway	NO	1,207,883	189,060	15.65
27	Poland	PL	3,030,384	467,033	15.41
28	Lithuania	LT	327,952	49,525	15.10
29	Estonia	EE	182,364	26,243	14.39
30	Cyprus	CY	115,415	16,038	13.90
31	Bulgaria	BG	368,235	49,724	13.50
32	Italy	IT	4,005,392	539,504	13.47
33	Montenegro	ME	56,044	7,498	13.38
34	Macedonia	MK	68,219	8,943	13.11
35	Latvia	LV	263,163	32,967	12.53
36	Faroe Islands	FO	13,226	1,654	12.51
37	Slovenia	SI	252,487	29,961	11.87
38	Greece	GR	888,354	104,360	11.75
39	Portugal	PT	870,579	102,083	11.73
40	Albania	AL	125,492	11,242	8.96
41	Ukraine (with Crimea)	UA	1,441,203	114,036	7.91
42	Georgia	GE	189,701	11,521	6.07

Continued on next page

Rank	Country	ISO2	Drivable ways	Tagged ways	Share (%)
43	Bosnia-Herzegovina	BA	214,701	10,729	5.00
44	Moldova	MD	188,319	9,175	4.87
45	Turkey	TR	1,902,558	59,063	3.10

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